

CAK2HAB3 - Dasar Kecerdasan Artifisial

Reasoning: First Order Logic

Lecturer Team





Five types of Logic

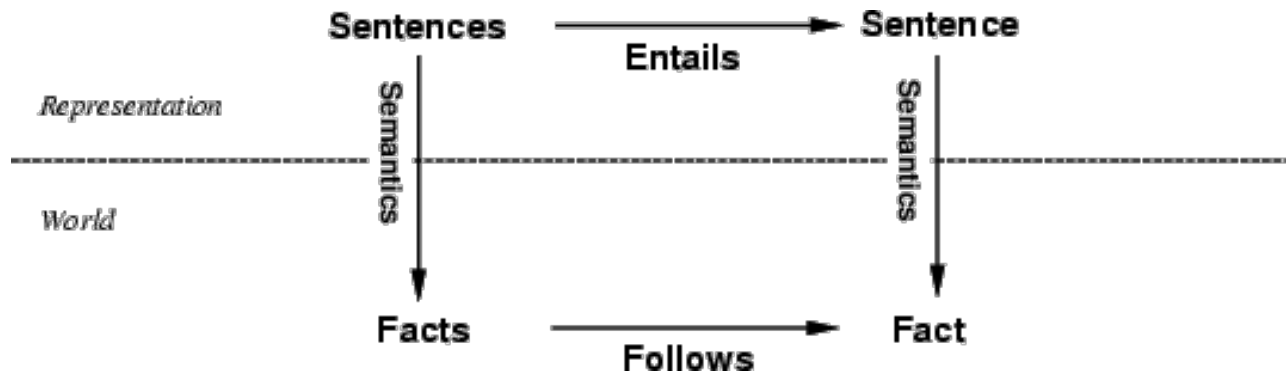
Type of logic	Reference to real world	Type of knowledge known by Agent
Propositional Logic	Facts	True / False / Unknown
First-order Logic	Facts, Object, Relation	True / False / Unknown
Temporal Logic	Facts, Object, Relation, Time	True / False / Unknown
Probability Theory	Facts	Degree of belief [0, 1] (certainty factor)
Fuzzy Logic	Truth values (degree of truth)	Confidence level [0, 1] (certainty factor)



Representation, reasoning, and logic

- ▶ The object of knowledge representation is to express knowledge in a computer-tractable form, so that agents can perform well.
- ▶ A knowledge representation language is defined by:
 - its syntax, which defines all possible sequences of symbols that constitute sentences of the language.
 - its semantics, which determines the facts in the world to which the sentences refer.

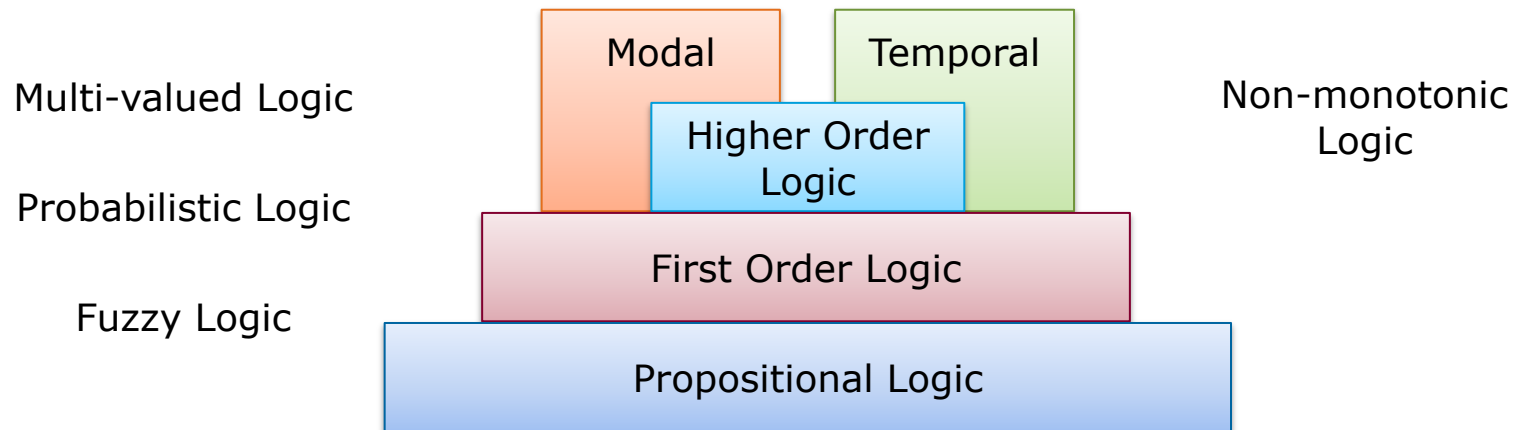
Sentences and Facts



- Semantics maps **sentences in logic** to **facts in the world**.
- The property of one fact following from another is mirrored by the property of one sentence being entailed by another.



Logic as a Knowledge-Representation Language





First Order Logic



First-Order Logic

- ▶ Propositional logic represents facts
 - Useful --- but has Limited Expressive Power
- ▶ First-order logic gives
 - Objects (*people, houses, colors, ...*)
 - Relations: how objects relate to each other (*brother of, bigger than, part of, ...*)
 - Properties: features of an object (*red, circle, ...*)
 - Functions: output an object, given others (*father of, best friend, ...*)

Sentence $\rightarrow$ *AtomicSentence*

| *Sentence Connective Sentence*

| *Quantifier Variable, ... Sentence*

| $\neg$ *Sentence*

| (*Sentence*)

AtomicSentence $\rightarrow$ *Predicate(Term,...)* | *Term = Term*

Term $\rightarrow$ *Function(Term,...)*

| *Constant*

| *Variable*

Connective $\rightarrow$ $\Rightarrow$ | $\wedge$ | $\vee$ | $\Leftrightarrow$

Quantifier $\rightarrow$ $\forall$ | $\exists$

Constant $\rightarrow$ *A* | *X₁* | *John* | ...

Variable $\rightarrow$ *a* | *x* | *s* | ...

Predicate $\rightarrow$ *Before* | *HasColor* | *Raining* | ...

Function $\rightarrow$ *MotherOf* | *LeftLegOf* | ...



Syntax and Semantics: Term

- ▶ Constant Symbols
 - *Book, Fred, Ali*, etc.
- ▶ Variables Symbols:
 - *x, y, a*, etc.
- ▶ Predicate Symbols :
 - *Brother(Fahmi, Galih) ; Mother(Ann, Sue) ;*
 - *Father(Luke, DarthVader) ;* etc.
- ▶ Function Symbols:
 - *ShoesOf(Caca) ; LeftLegOf(John) ;* etc



Syntax and Semantics: Sentences

▶ Atomic Sentences

- $Gives(Eko, Danu, ChocolateCake)$
- $BrotherOf(Fahmi) = Budi$
- $Siblings(SonOf(Fred), DaughterOf(Fred))$

▶ Complex Sentences

- $Brothers(Fahmi, Galih) \Rightarrow Gives(Fahmi, Galih, Chocolate)$
- $\neg ItsRaining(Today)$
- $Today(Monday) \vee Today(Tuesday)$



Universal Quantifiers

► “For all, For Every” : $\forall$

– $\forall x, Cat(x) \Rightarrow Mammal(x)$

– $\forall x, Child(x) \Rightarrow Likes(x, Candy)$

▪ This sentence is true iff all these sentence is true

– $Child(Andi) \Rightarrow Likes(Andi, Candy) \wedge Child(Anto) \Rightarrow Likes(Anto, Candy) \wedge Child(Budi) \Rightarrow Likes(Budi, Candy) \wedge \dots$



Universal Quantifiers

- ▶ Usually use $\Rightarrow$ with $\forall$
 - Common mistake to use $\wedge$
 - $\forall x, Student(x, Tel_U) \wedge Smart(x)$
 - Means everyone is a student of Tel-U and everyone is smart



Existential Quantifiers

- ▶ “There Exists”: $\exists$

- $\exists x, \text{Student}(x, \text{Telkom}) \wedge \text{Smart}(x)$
- $\exists x, \text{Mammal}(x) \wedge \text{Carnivore}(x)$

- ▶ Usually use $\wedge$ with $\exists$

- Common mistake to use $\Rightarrow$
 - $\exists x, \text{Child}(x) \Rightarrow \text{Likes}(x, \text{Candy})$



Properties of quantifiers

- ▶ $\forall x \forall y$ same as $\forall y \forall x$
- ▶ $\exists x \exists y$ same as $\exists y \exists x$
- ▶ $\exists x \forall y$ not the same as $\forall y \exists x$
 - $\forall y \exists x, LikesToEat(x, y)$
 - Every food in the world is favored by at least one person
 - $\exists x \forall y, LikesToEat(x, y)$
 - There is a person who loves every food in the world



Connections between $\forall$ and $\exists$

▶ $\forall x \neg Likes(x, IceCream)$ is equivalent to $\neg \exists x Likes(x, IceCream)$

▶ De Morgan Rule

$$\neg \forall x \neg P \equiv \neg \exists x P$$

$$\neg \forall x P \equiv \exists x \neg P$$

$$\neg \forall x P \equiv \neg \exists x \neg P$$

$$\neg \exists x P \equiv \neg \forall x \neg P$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$P \wedge Q \equiv \neg(\neg P \vee \neg Q)$$

$$P \vee Q \equiv \neg(\neg P \wedge \neg Q)$$



Inference in First-Order Logic

- ▶ Uses 7 rules from Propositional Logic
- ▶ With three additional rules for more complex case regarding to the quantifier
 - Universal Instantiation or Universal Elimination
 - Existential Instantiation or Existential Elimination
 - Existential Introduction or Existential Generalization



Example – Arm Dealer

- ▶ The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.
- ▶ Q: is Colonel West a Criminal?



Example – Arm Dealer – Objects and Relations

- ▶ The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.
- ▶ Object: *american(X)*, *criminal(X)*, *Weapon(Y)*, *Hostile(Z)*
- ▶ Relation: *sells(who,what,to)*, *owns(who,what)*, *enemy(who,to_who)*

Example – Arm Dealer – Clauses

- ▶ It is a crime for an American to sell weapons to hostile nations.
 - $american(X) \wedge weapon(Y) \wedge sells(X, Y, Z) \wedge hostile(Z) \Rightarrow criminal(X)$
- ▶ Missiles are weapon
 - $missile(X) \Rightarrow weapon(X)$
- ▶ An enemy of America counts as "hostile"
 - $enemy(X, america) \Rightarrow hostile(X)$

Example – Arm Dealer – Facts

- ▶ Nono ... has some missiles:
 - $owns(nono, m1) \wedge missile(m1)$
- ▶ all of its missiles were sold to it by Colonel West
 - $missile(X) \wedge owns(nono, X) \Rightarrow sells(west, X, nono)$
- ▶ West, who is American ...
 - $american(west)$

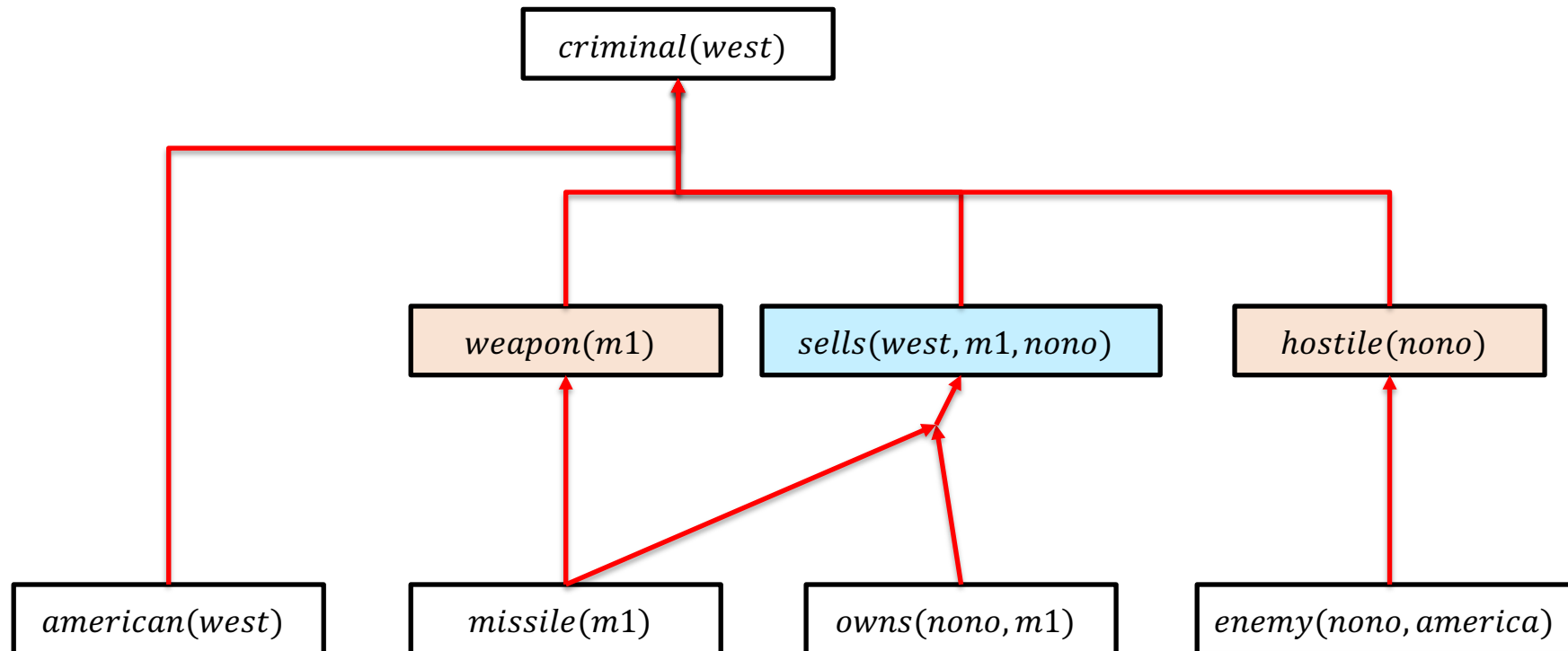


Example – Arm Dealer – Facts

- ▶ The country Nono, an enemy of America ...
 - *enemy(nono, america)*

- ▶ Question: is West a criminal?
 - *criminal(west)*

Example – Arm Dealer – Forward Chaining





Example

- 1. Lucy* is a professor
- 2. All professors are people.
- 3. John is the dean.
- 4. Deans are professors.
- 5. All professors consider the dean a friend or don't know him.
- 6. Everyone is a friend of someone.
- 7. People only criticize people that are not their friends.
- 8. Lucy criticized John .
- Question: IS JOHN NO FRIEND OF LUCY?



Example: Avunculate marriage

- ▶ Avunculate marriage is illegal and prohibited
 - marriage between an uncle and a niece or between an aunt and a nephew (second-degree relative).
- ▶ Case:
 - W married to A
 - W is biological daughter of B, where A and B is a twin
- ▶ Prove that A and W's marriage is invalid
 - Using FOL

Example: Avunculate marriage

‣ $\forall x, y \text{ NieceOf}(x, y) \wedge \text{Married}(x, y) \Rightarrow \neg \text{Valid}(\text{Married}(x, y))$ (1)

‣ $\text{Married}(W, A)$ (2)

‣ $\text{DaughterOf}(W, B)$ (3)

‣ $\text{Twins}(B, A)$ (4)

‣ $\forall x, y \text{ Twin}(x, y) \Rightarrow \text{Siblings}(x, y)$ (5)

‣ $\forall x, y, z \text{ DaughterOf}(x, y) \wedge \text{Siblings}(y, z) \Rightarrow \text{NieceOf}(x, z)$ (6)



Example: Avunculate marriage

- ▶ $Married(W, A) \wedge DaughterOf(W, B) \wedge Twins(B, A)$
- ▶ Universal Elimination of (5)
 - $\forall x, y \text{ } Twins(x, y) \Rightarrow Siblings(x, y)$
 - $Twins(B, A) \Rightarrow Siblings(B, A)$ (7)
- ▶ Modus Ponens of (4) and (7)
 - $Siblings(B, A)$ (8)



Example: Avunculate marriage

- ▶ $Married(W, A) \wedge DaughterOf(W, B) \wedge Twins(B, A) \wedge Siblings(B, A)$
- ▶ Universal Elimination of (6)
 - $\forall x, y, z \text{ DaughterOf}(x, y) \wedge Siblings(y, z) \Rightarrow NieceOf(x, z)$
 - $DaughterOf(W, B) \wedge Siblings(B, A) \Rightarrow NieceOf(W, A)$ (9)
- ▶ And-Introduction from (3) and (8)
 - $DaughterOf(W, B) \wedge Siblings(B, A)$ (10)
- ▶ Modus Ponens of (9) and (10)
 - $NieceOf(W, A)$ (11)

Example: Avunculate marriage

- ▶ $Married(W, A) \wedge DaughterOf(W, B) \wedge Twins(B, A) \wedge Siblings(B, A) \wedge NieceOf(W, A)$
- ▶ Universal Elimination of (1)
 - $\forall x, y \ NieceOf(x, y) \wedge Married(x, y) \Rightarrow \neg Valid(Married(x, y))$
 - $NieceOf(W, A) \wedge Married(W, A) \Rightarrow \neg Valid(Married(W, A))$ (12)
- ▶ And-Introduction from (11) and (2)
 - $NieceOf(W, A) \wedge Married(W, A)$ (13)
- ▶ Modus Ponens of (12) and (13)
 - $\neg Valid(Married(W, A))$ (14)

Question?





THANK YOU